HAT-P-36b

R-1605

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_200418` · created 2026-08-02T02:48:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458957.8862 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1558 +/- 0.0068
Transit depth (Rp/Rs)^2	2.43 +/- 0.21 [%]
Orbital Inclination (inc)	82.97 +/- 2.35
Airmass coefficient 1 (a1)	1.293 +/- 0.014
Airmass coefficient 2 (a2)	-0.1613 +/- 0.005
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	None
Optimal Aperture	4.32
Optimal Annulus	15.78
Transit Duration (day)	0.0874 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1558 ± 0.0068 vs 0.126 (deviation 4.15σ)
Duration fit vs published	0.0874 d vs 0.0925 d (deviation 0.51σ)
Mid-time O–C	8.5 min
Depth SNR	21.7
Residual scatter	1.11 %

Light curve

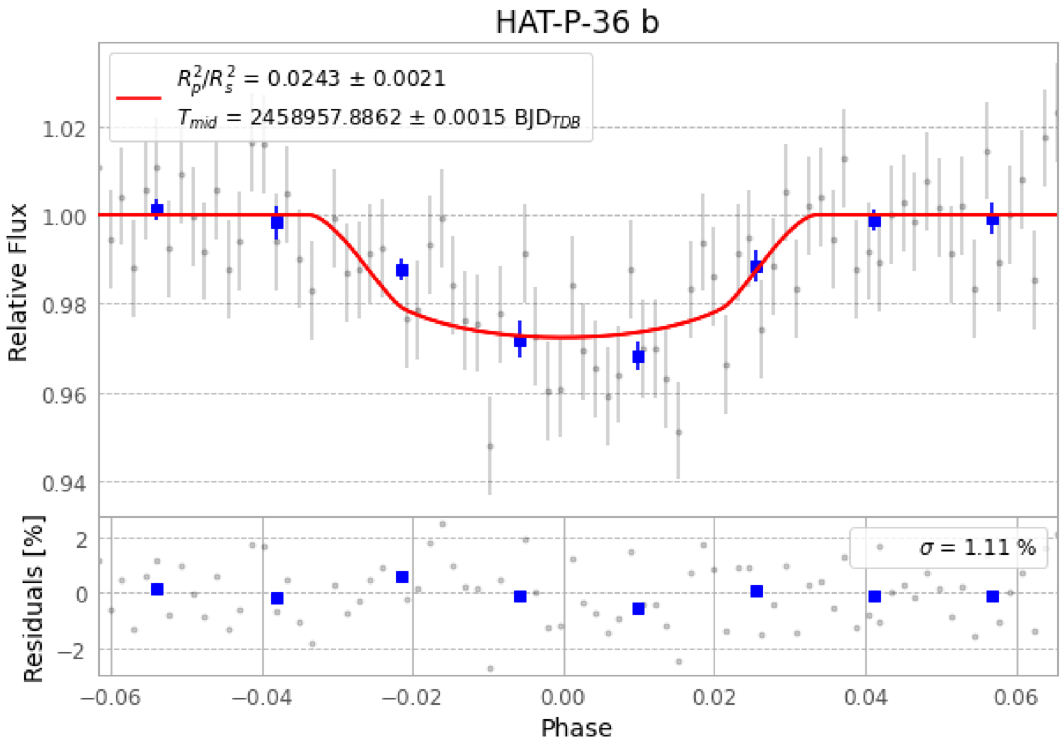

Figure 1 — FinalLightCurve_HAT-P-36 b_2020-04-18.png (SHA-256 9dd104f356770b8a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [468, 292], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a3522dee5c59cc6d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

82 FITS frames (54 MB), `HATP-362004180712 12.FITS` → `HATP-362004181115 12.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200418.log (e8abf5923274…), FinalLightCurve_HAT-P-36 b_2020-04-18.png (9dd104f35677…), FinalLightCurve_HAT-P-36 b_2020-04-18.csv (4a9da1876f90…)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)